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Finding Structure in Time

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Time underlies many interesting human behaviors. Thus, the question of how to represent time in connectionist models is very important. One approach is to represent time implicitly by its effects on processing rather than explicitly (as in a spatial representation). The current report develops a proposal along these lines first described by Jordan (1986) which involves the use of recurrent links in order to provide networks with a dynamic memory. In this approach, hidden unit patterns are fed back to themselves; the internal representations which develop thus reflect task demands in the context of prior internal states. A set of simulations is reported which range from relatively simple problems (temporal version of XOR) to discovering syntactic/semantic features for words. The networks are able to learn interesting internal representations which incorporate task demands with memory demands; indeed, in this approach the notion of memory is inextricably bound up with task processing. These representations reveal a rich structure, which allows them to be highly context-dependent, while also expressing generalizations across classes of items. These representations suggest a method for representing lexical categories and the type/token distinction.

INTRODUCTION

Time is clearly important in cognition. It is inextricably bound up with many behaviors (such as language) which express themselves as temporal sequences. Indeed, it is difficult to know how one might deal with such basic problems as goal-directed behavior, planning, or causation without some way of representing time.

The question of how to represent time might seem to arise as a special problem unique to parallel-processing models, if only because the parallel nature of computation appears to be at odds with the serial nature of tem-

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poral events. However, even within traditional (serial) frameworks, the representation of serial order and the interaction of a serial input or output with higher levels of representation presents challenges. For example, in models of motor activity, an important issue is whether the action plan is a literal specification of the output sequence, or whether the plan represents serial order in a more abstract manner (e.g., Fowler, 1977, 1980; Jordan & Rosenbaum, 1988; Kelso, Saltzman, & Tuller, 1986; Lashley, 1951; MacNeilage, 1970; Saltzman & Kelso, 1987). Linguistic theoreticians have perhaps tended to be less concerned with the representation and processing of the temporal aspects to utterances (assuming, for instance, that all the information in an utterance is somehow made available simultaneously in a syntactic tree); but the research in natural language parsing suggests that the problem is not trivially solved (e.g., Frazier & Fodor, 1978; Marcus, 1980). Thus, what is one of the most elementary facts about much of human activity—that it has temporal extend—is sometimes ignored and is often problematic.

In parallel distributed processing models, the processing of sequential inputs has been accomplished in several ways. The most common solution is to attempt to "parallelize time" by giving it a spatial representation. However, there are problems with this approach, and it is ultimately not a good solution. A better approach would be to represent time implicitly rather than explicitly. That is, we represent time by the effect it has on processing and not as an additional dimension of the input.

This article describes the results of pursuing this approach, with particular emphasis on problems that are relevant to natural language processing. The approach taken is rather simple, but the results are sometimes complex and unexpected. Indeed, it seems that the solution to the problem of time may interact with other problems for connectionist architectures, including the problem of symbolic representation and how connectionist representations encode structure. The current approach supports the notion outlined by Van Gelder (in press) (see also, Elman, 1989; Smolensky, 1987, 1988), that connectionist representations may have a functional compositionality without being syntactically compositional.

The first section briefly describes some of the problems that arise when time is represented externally as a spatial dimension. The second section describes the approach used in this work. The major portion of this article presents the results of applying this new architecture to a diverse set of problems. These problems range in complexity from a temporal version of the Exclusive-OR function to the discovery of syntactic/semantic categories in natural language data.

THE PROBLEM WITH TIME

One obvious way of dealing with patterns that have a temporal extent is to represent time explicitly by associating the serial order of the pattern with

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the dimensionality of the pattern vector. The first temporal event is represented by the first element in the pattern vector, the second temporal event is represented by the second position in the pattern vector, and so on. The entire pattern vector is processed in parallel by the model. This approach has been used in a variety of models (e.g., Cottrell, Munro, & Zipser, 1987; Elman & Zipser, 1988; Hanson & Kegl, 1987).

There are several drawbacks to this approach, which basically uses a spatial metaphor for time. First, it requires that there be some interface with the world, which buffers the input, so that it can be presented all at once. It is not clear that biological systems make use of such shift registers. There are also logical problems: How should a system know when a buffer's contents should be examined?

Second, the shift register imposes a rigid limit on the duration of patterns (since the input layer must provide for the longest possible pattern), and furthermore, suggests that all input vectors be the same length. These problems are particularly troublesome in domains such as language, where one would like comparable representations for patterns that are of variable length. This is as true of the basic units of speech (phonetic segments) as it is of sentences.

Finally, and most seriously, such an approach does not easily distinguish relative temporal position from absolute temporal position. For example, consider the following two vectors.

[0 1 1 1 0 0 0 0 0]
[0 0 0 1 1 1 0 0 0]

These two vectors appear to be instances of the same basic pattern, but displaced in space (or time, if these are given a temporal interpretation). However, as the geometric interpretation of these vectors makes clear, the two patterns are in fact quite dissimilar and spatially distant.[1] PDP models can, of course, be trained to treat these two patterns as similar. But the similarity is a consequence of an external teacher and not of the similarity structure of the patterns themselves, and the desired similarity does not generalize to novel patterns. This shortcoming is serious if one is interested in patterns in which the relative temporal structure is preserved in the face of absolute temporal displacements.

What one would like is a representation of time that is richer and does not have these problems. In what follows here, a simple architecture is described, which has a number of desirable temporal properties, and has yielded interesting results.

[1] The reader may more easily be convinced of this by comparing the locations of the vectors **[1 0 0]**, **[0 1 0]**, and **[0 0 1]** in 3-space. Although these patterns might be considered "temporally displaced" versions of the same basic pattern, the vectors are very different.

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NETWORKS WITH MEMORY

The spatial representation of time described above treats time as an explicit part of the input. There is another, very different possibility: Allow time to be represented by the effect it has on processing. This means giving the processing system dynamic properties that are responsive to temporal sequences. In short, the network must be given memory.

There are many ways in which this can be accomplished, and a number of interesting proposals have appeared in the literature (e.g., Jordan, 1986; Pineda, 1988; Stornetta, Hogg, & Huberman, 1987; Tank & Hopfield, 1987; Waibel, Hanazawa, Hinton, Shikano, & Lang, 1987; Watrous & Shastri, 1987; Williams & Zipser, 1988). One of the most promising was suggested by Jordan (1986). Jordan described a network (shown in Figure 1) containing recurrent connections that were used to associate a static pattern (a "Plan") with a serially ordered output pattern (a sequence of "Actions"). The recurrent connections allow the network's hidden units to see its own previous output, so that the subsequent behavior can be shaped by previous responses. These recurrent connections are what give the network memory.

This approach can be modified in the following way. Suppose a network (shown in Figure 2) is augmented at the input level by additional units; call these *Context Units.* These units are also "hidden" in the sense that they interact exclusively with other nodes internal to the network, and not the outside world.

Imagine that there is a sequential input to be processed, and some clock which regulates presentation of the input to the network. Processing would then consist of the following sequence of events. At time t, the input units receive the first input in the sequence. Each unit might be a single scalar value or a vector, depending upon the nature of the problem. The context units are initially set to 0.5.[2] Both the input units and context units activate the hidden units; the hidden units then feed forward to activate the output units. The hidden units also feed back to activate the context units. This constitutes the forward activation. Depending upon the task, there may or may not be a learning phase in this time cycle. If so, the output is compared with a teacher input, and back propagation of error (Rumelhart, Hinton, & Williams, 1986) is used to adjust connection strengths incrementally. Recurrent connections are fixed at 1.0 and are not subject to adjustment.[3] At the next time step, $t+1$, the above sequence is repeated. This time the context

[2] The activation function used here bounds values between 0.0 and 1.0.

[3] A little more detail is in order about the connections between the context units and hidden units. In the networks used here, there were one-for-one connections between each hidden unit and each context unit. This implies that there are an equal number of context and hidden units. The upward connections between the context units and the hidden units were fully distributed, such that each context unit activates all the hidden units.

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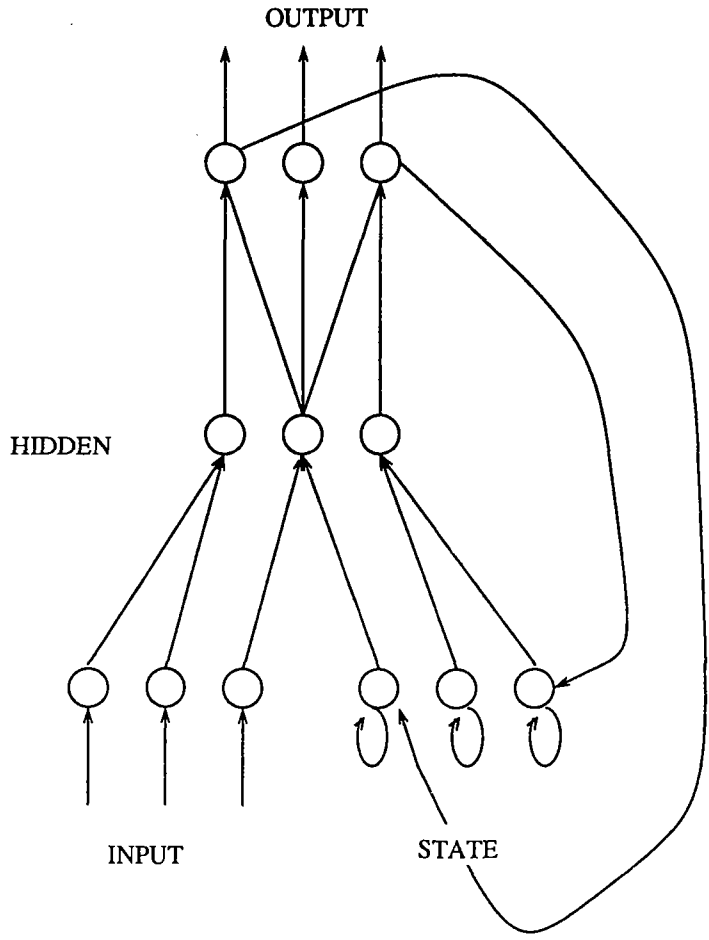


Figure 1. Architecture used by Jordan (1986). Connections from output to state units are one-for-one, with a fixed weight of 1.0 Not all connections are shown.

units contain values which are exactly the hidden unit values at time *t*. These context units thus provide the network with memory.

Internal Representation of Time. In feed forward networks employing hidden units and a learning algorithm, the hidden units develop internal representations for the input patterns that recode those patterns in a way which enables the network to produce the correct output for a given input. In the present architecture, the context units remember the previous internal state. Thus, the hidden units have the task of mapping both an external

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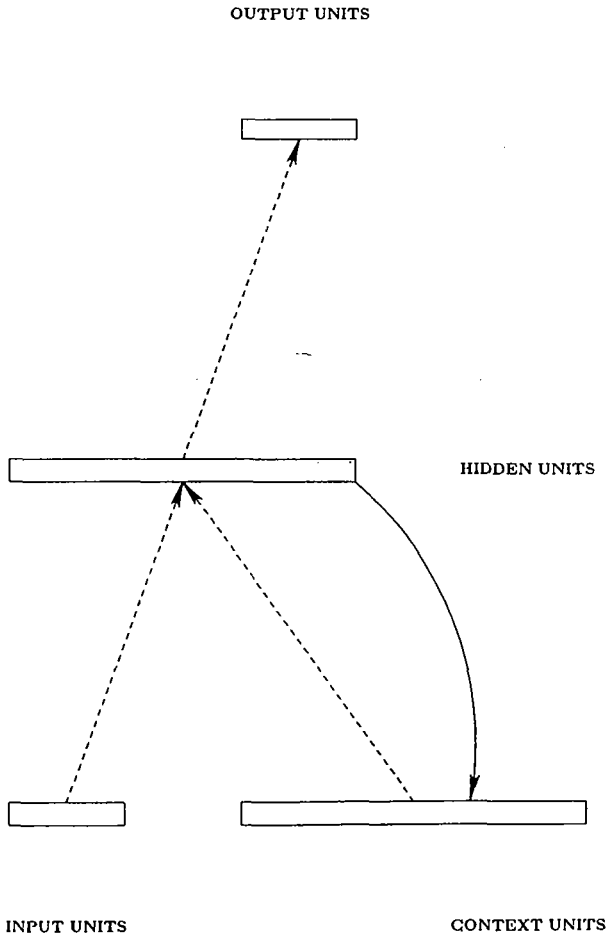


Figure 2. A simple recurrent network in which activations are copied from hidden layer to context layer on a one-for-one basis, with fixed weight of 1.0. Dotted lines represent trainable connections.

input, and also the previous internal state of some desired output. Because the patterns on the hidden units are saved as context, the hidden units must accomplish this mapping and at the same time develop representations which are useful encodings of the temporal properties of the sequential input. Thus, the internal representations that develop are sensitive to temporal context; the effect of time is implicit in these internal states. Note, however, that these representations of temporal context need not be literal. They represent a memory which is highly task- and stimulus-dependent.

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Consider now the results of applying this architecture to a number of problems that involve processing of inputs which are naturally presented in sequence.

EXCLUSIVE-OR

The Exclusive-Or (XOR) function has been of interest because it cannot be learned by a simple two-layer network. Instead, it requires at least three layers. The XOR is usually presented as a problem involving 2-bit input vectors (00, 11, 01, 10) yielding 1-bit output vectors (0, 0, 1, 1, respectively).

This problem can be translated into a temporal domain in several ways. One version involves constructing a sequence of 1-bit inputs by presenting the 2-bit inputs one bit at a time (i.e., in 2 time steps), followed by the 1-bit output; then continuing with another input/output pair chosen at random. A sample input might be:

1 0 1 0 0 0 0 1 1 1 1 0 1 0 1...

Here, the first and second bits are XOR-ed to produce the third; the fourth and fifth are XOR-ed to give the sixth; and so on. The inputs are concatenated and presented as an unbroken sequence.

In the current version of the XOR problem, the input consisted of a sequence of 3,000 bits constructed in this manner. This input stream was presented to the network shown in Figure 2 (with 1 input unit, 2 hidden units, 1 output unit, and 2 context units), one bit at a time. The task of the network was, at each point in time, to predict the next bit in the sequence. That is, given the input sequence shown, where one bit at a time is presented, the correct output at corresponding points in time is shown below.

input: **1 0 1 0 0 0 0 1 1 1 1 0 1 0 1...**
output: **0 1 0 0 0 0 1 1 1 1 0 1 0 1 ?...**

Recall that the actual input to the hidden layer consists of the input shown above, as well as a copy of the hidden unit activations from the previous cycle. The prediction is thus based not just on input from the world, but also on the network's previous state (which is continuously passed back to itself on each cycle).

Notice that, given the temporal structure of this sequence, it is only sometimes possible to predict the next item correctly. When the network has received the first bit—**1** in the example above—there is a 50% chance that the next bit will be a **1** (or a **0**). When the network receives the second bit (**0**), however, it should then be possible to predict that the third will be the XOR, **1**. When the fourth bit is presented, the fifth is not predictable. But from the fifth bit, the sixth can be predicted, and so on.

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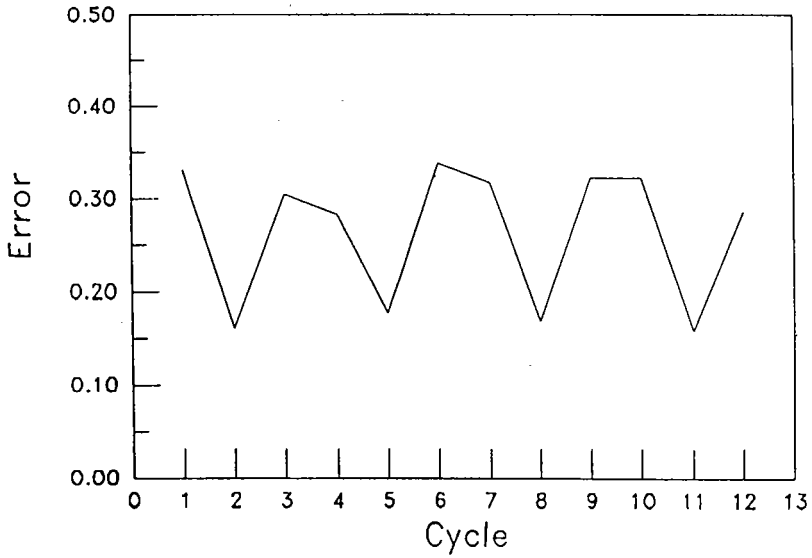


Figure 3. Graph of root mean squared error over 12 consecutive inputs in sequential XOR task. Data points are averaged over 1200 trials.

In fact, after 600 passes through a 3,000-bit sequence constructed in this way, the network's ability to predict the sequential input closely follows the above schedule. This can be seen by looking at the sum squared error in the output prediction at successive points in the input. The error signal provides a useful guide as to when the network recognized a temporal sequence, because at such moments its outputs exhibit low error. Figure 3 contains a plot of the sum squared error over 12 time steps (averaged over 1,200 cycles). The error drops at those points in the sequence where a correct prediction is possible; at other points, the error is high. This is an indication that the network has learned something about the temporal structure of the input, and is able to use previous context and current input to make predictions about future input. The network, in fact, attempts to use the XOR rule at all points in time; this fact is obscured by the averaging of error, which is done for Figure 3. If one looks at the output activations, it is apparent from the nature of the errors that the network predicts successive inputs to be the XOR of the previous two. This is guaranteed to be successful every third bit, and will sometimes, fortuitously, also result in correct predictions at other times.

It is interesting that the solution to the temporal version of XOR is somewhat different than the static version of the same problem. In a network with two hidden units, one unit is highly activated when the input sequence is a series of identical elements (all 1s or 0s), whereas the other unit is highly

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activated when the input elements alternate. Another way of viewing this is that the network develops units which are sensitive to high- and low-frequency inputs. This is a different solution than is found with feed-forward networks and simultaneously presented inputs. This suggests that problems may change their nature when cast in a temporal form. It is not clear that the solution will be easier or more difficult in this form, but it is an important lesson to realize that the solution may be different.

In this simulation, the prediction task has been used in a way that is somewhat analogous to auto-association. Auto-association is a useful technique for discovering the intrinsic structure possessed by a set of patterns. This occurs because the network must transform the patterns into more compact representations; it generally does so by exploiting redundancies in the patterns. Finding these redundancies can be of interest because of what they reveal about the similarity structure of the data set (cf. Cottrell et al. 1987; Elman & Zipser, 1988).

In this simulation, the goal is to find the temporal structure of the XOR sequence. Simple auto-association would not work, since the task of simply reproducing the input at all points in time is trivially solvable and does not require sensitivity to sequential patterns. The prediction task is useful because its solution requires that the network be sensitive to temporal structure.

STRUCTURE IN LETTER SEQUENCES

One question which might be asked is whether the memory capacity of the network architecture employed here is sufficient to detect more complex sequential patterns than the XOR. The XOR pattern is simple in several respects. It involves single-bit inputs, requires a memory which extends only one bit back in time, and has only four different input patterns. More challenging inputs would require multi-bit inputs of greater temporal extent, and a larger inventory of possible sequences. Variability in the duration of a pattern might also complicate the problem.

An input sequence was devised which was intended to provide just these sorts of complications. The sequence was composed of six different 6-bit binary vectors. Although the vectors were not derived from real speech, one might think of them as representing speech sounds, with the six dimensions of the vector corresponding to articulatory features. Table 1 shows the vector for each of the six letters.

The sequence was formed in two steps. First, the three consonants (b, d, g) were combined in random order to obtain a 1,000-letter sequence. Then, each consonant was replaced using the rules

b → ba
d → dii
g → guuu

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TABLE 1
Vector Definitions of Alphabet

	Consonant	Vowel	Interrupted	High	Back	Voiced
b	[1	0	1	0	0	1]
d	[1	0	1	1	0	1]
g	[1	0	1	0	1	1]
a	[0	1	0	0	1	1]
i	[0	1	0	1	0	1]
u	[0	1	0	1	1	1]

Thus, an initial sequence of the form **dbgbddg**... gave rise to the final sequence **diibaguuubadiidiiguuu**... (each letter being represented by one of the above 6-bit vectors). The sequence was semi-random; consonants occurred randomly, but following a given consonant, the identity and number of following vowels was regular.

The basic network used in the XOR simulation was expanded to provide for the 6-bit input vectors; there were 6 input units, 20 hidden units, 6 output units, and 20 context units.

The training regimen involved presenting each 6-bit input vector, one at a time, in sequence. The task for the network was to predict the next input. (The sequence wrapped around, that the first pattern was presented after the last.) The network was trained on 200 passes through the sequence. It was then tested on another sequence that obeyed the same regularities, but created from a different initial randomizaiton.

The error signal for part of this testing phase is shown in Figure 4. Target outputs are shown in parenthesis, and the graph plots the corresponding error for each prediction. It is obvious that the error oscillates markedly; at some points in time, the prediction is correct (and error is low), while at other points in time, the ability to predict correctly is quite poor. More precisely, error tends to be high when predicting consonants, and low when predicting vowels.

Given the nature of the sequence, this behavior is sensible. The consonants were ordered randomly, but the vowels were not. Once the network has received a consonant as input, it can predict the identity of the following vowel. Indeed, it can do more; it knows how many tokens of the vowel to expect. At the end of the vowel sequence it has no way to predict the next consonant; at these points in time, the error is high.

This global error pattern does not tell the whole story, however. Remember that the input patterns (which are also the patterns the network is trying to predict) are bit vectors. The error shown in Figure 4 is the sum squared error over all 6 bits. Examine the error on a bit-by-bit basis; a graph of the error for bits [1] and [4] (over 20 time steps) is shown in Figure 5. There is a

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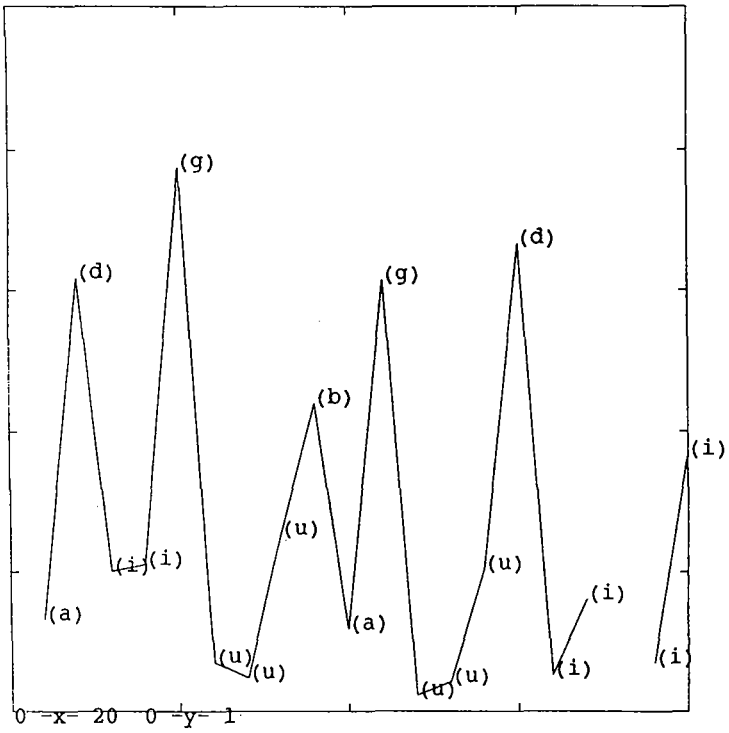


Figure 4. Graph of root mean squared error in letter prediction task. Labels indicate the correct output prediction at each point in time. Error is computed over the entire output vector.

striking difference in the error patterns. Error on predicting the first bit is consistently lower than error for the fourth bit, and at all points in time. Why should this be so?

The first bit corresponds to the features **Consonant**; the fourth bit corresponds to the feature **High.** It happens that while all consonants have the same value for the feature **Consonant,** they differ for **High.** The network has learned which vowels follow which consonants; this is why error on vowels is low. It has also learned how many vowels follow each consonant. An interesting corollary is that the network also knows how soon to expect the next consonant. The network cannot know *which* consonant, but it can predict correctly that a consonant follows. This is why the bit patterns for **Consonant** show low error, and the bit patterns for **High** show high error. (It is this behavior which requires the use of context units; a simple feed-forward network could learn the transitional probabilities from one input to the next, but could not learn patterns that span more than two inputs.)

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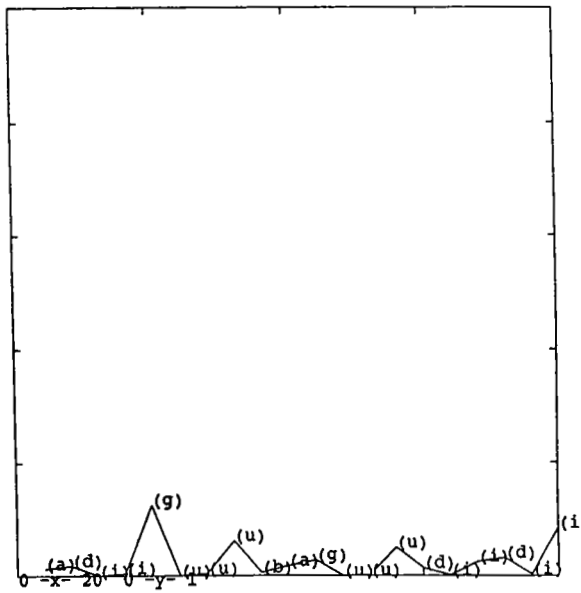


Figure 5 (a). Graph of root mean squared error in letter prediction task. Error is computed on bit 1, representing the feature CONSONANTAL.

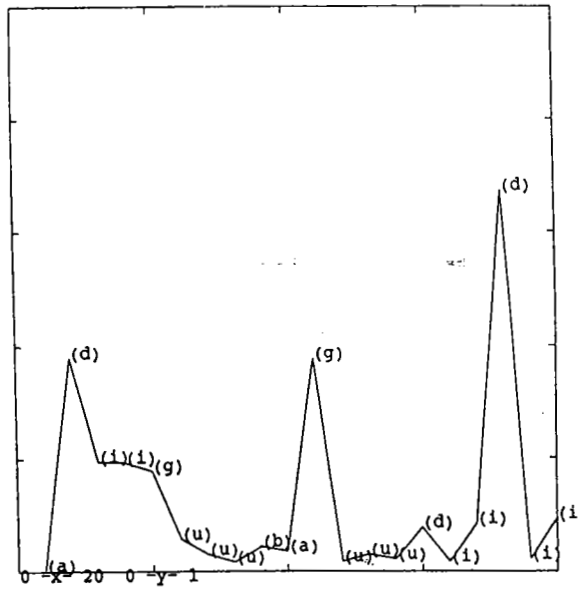


Figure 5 (b). Graph of root mean squared error in letter prediction task. Error is computed on bit 4, representing the feature HIGH.

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This simulation demonstrates an interesting point. This input sequence was in some ways more complex than the XOR input. The serial patterns are longer in duration; they are of variable length so that a prediction depends upon a variable mount of temporal context; and each input consists of a 6-bit rather than a 1-bit vector. One might have reasonably thought that the more extended sequential dependencies of these patterns would exceed the temporal processing capacity of the network. But almost the opposite is true. The fact that there are subregularities (at the level of individual bit patterns) enables the network to make partial predictions, even in cases where the complete prediction is not possible. All of this is dependent upon the fact that the input is structured, of course. The lesson seems to be that more extended sequential dependencies may not necessarily be more difficult to learn. If the dependencies are structured, that structure may make learning easier and not harder.

DISCOVERING THE NOTION "WORD"

It is taken for granted that learning a language involves (among many other things) learning the sounds of that language, as well as the morphemes and words. Many theories of acquisition depend crucially upon such primitive types as word, or morpheme, or more abstract categories as noun, verb, or phrase (e.g., Berwick & Weinberg, 1984; Pinker, 1984). Rarely is it asked how a language learner knows when to begin or why these entities exist. These notions are often assumed to be innate.

Yet, in fact, there is considerable debate among linguists and psycholinguists about what representations are used in language. Although it is commonplace to speak of basic units such as "phoneme," "morpheme," and "word," these constructs have no clear and uncontroversial definition. Moreover, the commitment to such distinct levels of representation leaves a troubling residue of entities that appear to lie between the levels. For instance, in many languages, there are sound/meaning correspondences which lie between the phoneme and the morpheme (i.e., sound symbolism). Even the concept "word" is not as straightforward as one might think (cf. Greenberg, 1963; Lehman, 1962). In English, for instance, there is no consistently definable distinction among words (e.g., "apple"), compounds ("apple pie") and phrases ("Library of Congress" or "man in the street"). Furthermore, languages differ dramatically in what they treat as words. In polysynthetic languages (e.g., Eskimo), what would be called words more nearly resemble what the English speaker would call phrases or entire sentences.

Thus, the most fundamental concepts of linguistic analysis have a fluidity, which at the very least, suggests an important role for learning; and the exact form of the those concepts remains an open and important question.

In PDP networks, representational form and representational content often can be learned simultaneously. Moreover, the representations which

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result have many of the flexible and graded characteristics noted above. Therefore, one can ask whether the notion "word" (or something which maps on to this concept) could emerge as a consequence of learning the sequential structure of letter sequences that form words and sentences (but in which word boundaries are not marked).

Imagine then, another version of the previous task, in which the latter sequences form real words, and the words form sentences. The input will consist of the individual letters (imagine these as analogous to speech sounds, while recognizing that the orthographic input is vastly simpler than acoustic input would be). The letters will be presented in sequence, one at a time, with no breaks between the letters in a word, and no breaks between the words of different sentences.

Such a sequence was created using a sentence-generating program and a lexicon of 15 words.[4] The program generated 200 sentences of varying length, from four to nine words. The sentences were concatenated, forming a stream of 1,270 words. Next, the words were broken into their letter parts, yielding 4,963 letters. Finally, each letter in each word was converted into a 5-bit random vector.

The result was a stream of 4,963 separate 5-bit vectors, one for each letter. These vectors were the input and were presented one at a time. The task at each point in time was to predict the next letter. A fragment of the input and desired output is shown in Table 2.

A network with 5 input units, 20 hidden units, 5 output units, and 20 context units was trained on 10 complete presentations of the sequence. The error was relatively high at this point; the sequence was sufficiently random that it would be difficult to obtain very low error without memorizing the entire sequence (which would have required far more than 10 presentations).

Nonetheless, a graph of error over time reveals an interesting pattern. A portion of the error is plotted in Figure 6; each data point is marked with the letter that should be predicted at that point in time. Notice that at the onset of each new word, the error is high. As more of the word is received the error declines, since the sequence is increasingly predictable.

The error provides a good clue as to what the recurring sequences in the input are, and these correlate highly with words. The information is not categorical, however. The error reflects statistics of co-occurrence, and these are graded. Thus, while it is possible to determine, more or less, what sequences constitute words (those sequences bounded by high error), the criteria for boundaries are relative. This leads to ambiguities, as in the case of the *y* in *they* (see Figure 6); it could also lead to the misidentification of

[4] The program used was a simplified version of the program described in greater detail in the next simulation.

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TABLE 2
Fragment of Training Sequence for Letters-in-Words Simulation

Input		Output	
01101	(*m*)	00001	(*a*)
00001	(*a*)	01110	(*n*)
01110	(*n*)	11001	(*y*)
11001	(*y*)	11001	(*y*)
11001	(*y*)	00101	(*e*)
00101	(*e*)	00001	(*a*)
00001	(*a*)	10010	(*r*)
10010	(*r*)	10011	(*s*)
10011	(*s*)	00001	(*a*)
00001	(*a*)	00111	(*g*)
00111	(*g*)	01111	(*o*)
01111	(*o*)	00001	(*a*)
00001	(*a*)	00010	(*b*)
00010	(*b*)	01111	(*o*)
01111	(*o*)	11001	(*y*)
11001	(*y*)	00001	(*a*)
00001	(*a*)	01110	(*n*)
01110	(*n*)	00100	(*d*)
00100	(*d*)	00111	(*g*)
00111	(*g*)	01001	(*i*)
01001	(*i*)	10010	(*r*)
10010	(*r*)	01100	(*l*)
01100	(*l*)	01100	(*i*)
11001	(*i*)		

common sequences that incorporate more than one word, but which co-occur frequently enough to be treated as a quasi-unit. This is the sort of behavior observed in children, who at early stages of language acquisition may treat idioms and other formulaic phrases as fixed lexical items (MacWhinney, 1978).

This simulation should not be taken as a model of word acquisition. While listeners are clearly able to make predictions based upon partial input (Grosjean, 1980; Marslen-Wilson & Tyler, 1980; Salasoo & Pisoni, 1985), prediction is not the major goal of the language learner. Furthermore, the co-occurrence of sounds is only part of what identifies a word as such. The environment in which those sounds are uttered, and the linguistic context, are equally critical in establishing the coherence of the sound sequence and associating it with meaning. This simulation focuses only on a limited part of the information available to the language learner. The simulation makes the simple point that there is information in the signal that could serve as a cue to the boundaries of linguistic units which must be learned, and it demonstrates the ability of simple recurrent nerworks to extract this information.

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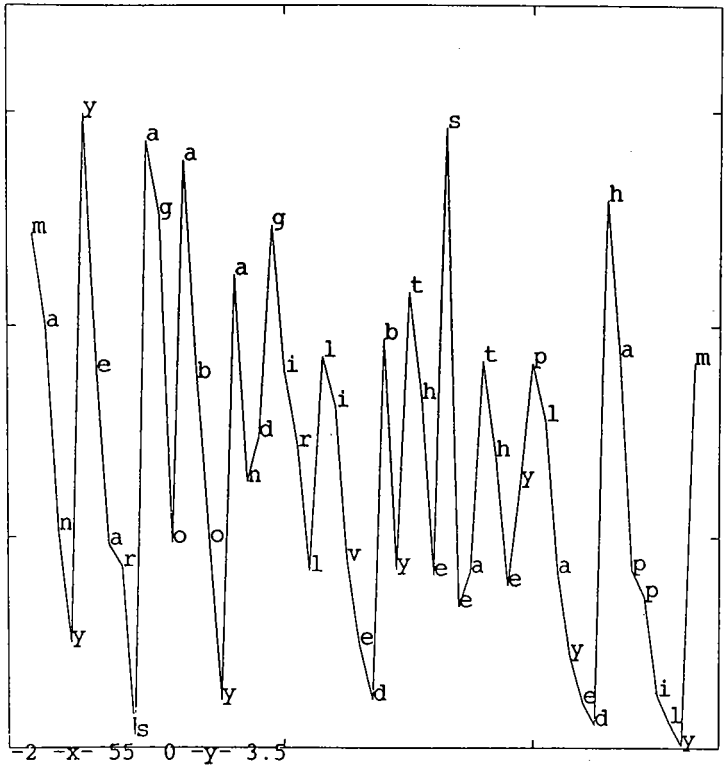


Figure 6. Graph of root mean squared error in letter-in-word precition task.

DISCOVERING LEXICAL CLASSES FROM WORD ORDER

Consider now another problem which arises in the context of word sequences. The order of words in sentences reflects a number of constraints. In languages such as English (so-called "fixed word-order" languages), the order is tightly constrained. In many other languages (the "free word-order" languages), there are more options as to word order (but even here the order is not free in the sense of random). Syntactic structure, selective restrictions, subcategorization, and discourse considerations are among the many factors which join together to fix the order in which words occur. Thus, the sequential order of words in sentences is neither simple, nor is it determined by a single cause. In addition, it has been argued that generalizations about word order cannot be accounted for solely in terms of linear order (Chomsky, 1957, 1965). Rather, there is an abstract structure which underlies the surface strings and it is this structure which provides a more insightful basis for understanding the constraints on word order.

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While it is undoubtedly true that the surface order of words does not provide the most insightful basis for generalizations about word order, it is also true that from the point of view of the listener, the surface order is the only visible (or audible) part. Whatever the abstract underlying structure be, it is cued by the surface forms, and therefore, that structure is implicit in them.

In the previous simulation, it was demonstrated that a network was able to learn the temporal structure of letter sequences. The order of letters in that simulation, however, can be given with a small set of relatively simple rules.[5] The rules for determining word order in English, on the other hand, will be complex and numerous. Traditional accounts of word order generally invoke symbolic processing systems to express abstract structural relationships. One might, therefore, easily believe that there is a qualitative difference in the nature of the computation needed for the last simulation, which is required to predict the word order of English sentences. Knowledge of word order might require symbolic representations that are beyond the capacity of (apparently) nonsymbolic PDP systems. Furthermore, while it is true, as pointed out above, that the surface strings may be cues to abstract structure, considerable innate knowledge may be required in order to reconstruct the abstract structure from the surface strings. It is, therefore, an interesting question to ask whether a network can learn any aspects of that underlying abstract structure.

Simple Sentences

As a first step, a somewhat modest experiment was undertaken. A sentence generator program was used to construct a set of short (two- and three-word) utterances. Thirteen classes of nouns and verbs were chosen; these are listed in Table 3. Examples of each category are given; it will be noticed that instances of some categories (e.g., VERB-DESTROY) may be included in others (e.g., VERB-TRAN). There were 29 different lexical items.

The generator program used these categories and the 15 sentence templates given in Table 4 to create 10,000 random two- and three-word sentence frames. Each sentence frame was then filled in by randomly selecting one of the possible words appropriate to each category. Each word was replaced by a randomly assigned 31-bit vector in which each word was represented by a different bit. Whenever the word was present, that bit was flipped on. Two extra bits were reserved for later simulations. This encoding scheme guaranteed that each vector was orthogonal to every other vector and reflected nothing about the form class or meaning of the words. Finally, the 27,534 word vectors in the 10,000 sentences were concatenated, so that an input stream of 27,534 31-bit vectors was created. Each word vector was distinct,

[5] In the worst case, each word constitutes a rule. Hopefully, networks will learn that recurring orthographic regularities provide additional and more general constraints (cf. Sejnowski & Rosenberg, 1987).

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TABLE 3
Categories of Lexical Items Used in Sentence Simulation

Category	Examples
NOUN-HUM	man, woman
NOUN-ANIM	cat, mouse
NOUN-INANIM	book, rock
NOUN-AGRESS	dragon, monster
NOUN-FRAG	glass, plate
NOUN-FOOD	cookie, bread
VERB-INTRAN	think, sleep
VERB-TRAN	see, chase
VERB-AGPAT	move, break
VERB-PERCEPT	smell, see
VERB-DESTROY	break, smash
VERB-EAT	eat

TABLE 4
Templates for Sentence Generator

WORD 1	WORD 2	WORD 3
NOUN-HUM	VERB-EAT	NOUN-FOOD
NOUN-HUM	VERB-PERCEPT	NOUN-INANIM
NOUN-HUM	VERB-DESTROY	NOUN-FRAG
NOUN-HUM	VERB-INTRAN	
NOUN-HUM	VERB-TRAN	NOUN-HUM
NOUN-HUM	VERB-AGPAT	NOUN-INANIM
NOUN-HUM	VERB-AGPAT	
NOUN-ANIM	VERB-EAT	NOUN-FOOD
NOUN-ANIM	VERB-TRAN	NOUN-ANIM
NOUN-ANIM	VERB-AGPAT	NOUN-INANIM
NOUN-ANIM	VERB-AGPAT	
NOUN-INANIM	VERB-AGPAT	
NOUN-AGRESS	VERB-DESTROY	NOUN-FRAG
NOUN-AGRESS	VERB-EAT	NOUN-HUM
NOUN-AGRESS	VERB-EAT	NOUN-ANIM
NOUN-AGRESS	VERB-EAT	NOUN-FOOD

but there were no breaks between successive sentences. A fragment of the input stream is shown in Column 1 of Table 5, with the English gloss for each vector in parentheses. The desired output is given in Column 2.

For this simulation a network similar to that in the first simulation was used, except that the input layer and output layers contained 31 nodes each, and the hidden and context layers contained 150 nodes each.

The task given to the network was to learn to predict the order of successive words. The training strategy was as follows. The sequence of 27,354

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TABLE 5
Fragment of Training Sequences for Sentence Simulation

Input	Output
00000000000000000000000000010 (*woman*)	00000000000000000000000000010000 (*smash*)
00000000000000000000000000010000 (*smash*)	00000000000000000000001000000000 (*plate*)
00000000000000000000001000000000 (*plate*)	00000100000000000000000000000000 (*cat*)
00000100000000000000000000000000 (*cat*)	00000000000000000000100000000000 (*move*)
00000000000000000000100000000000 (*move*)	00000000000000000100000000000000 (*man*)
00000000000000000100000000000000 (*man*)	00010000000000000000000000000000 (*break*)
00010000000000000000000000000000 (*break*)	00001000000000000000000000000000 (*car*)
00001000000000000000000000000000 (*car*)	01000000000000000000000000000000 (*boy*)
01000000000000000000000000000000 (*boy*)	00000000000000000000100000000000 (*move*)
00000000000000000000100000000000 (*move*)	00000000000010000000000000000000 (*girl*)
00000000000010000000000000000000 (*girl*)	00000000001000000000000000000000 (*eat*)
00000000001000000000000000000000 (*eat*)	00100000000000000000000000000000 (*bread*)
00100000000000000000000000000000 (*bread*)	00000000100000000000000000000000 (*dog*)
00000000100000000000000000000000 (*dog*)	00000000000000000000100000000000 (*move*)
00000000000000000000100000000000 (*move*)	00000000000000000001000000000000 (*mouse*)
00000000000000000001000000000000 (*mouse*)	00000000000000000001000000000000 (*mouse*)
00000000000000000001000000000000 (*mouse*)	00000000000000000000100000000000 (*move*)
00000000000000000000100000000000 (*move*)	10000000000000000000000000000000 (*book*)
10000000000000000000000000000000 (*book*)	00000000000000001000000000000000 (*lion*)

31-bit vectors formed an input sequence. Each word in the sequence was input, one at a time, in order. The task on each input cycle was to predict the 31-bit vector corresponding to the next word in the sequence. At the end of the 27,534 word sequence, the process began again, without a break, starting with the first word. The training continued in this manner until the network had experienced six complete passes through the sequence.

Measuring the performance of the network in this simulation is not straightforward. RMS error after training dropped to 0.88. When output vectors are as sparse as those used in this simulation (only 1 out of 31 bits turned on), the network quickly learns to turn off all the output units, which drops error from the initial random value of ~ 15.5 to 1.0. In this light, a final error of 0.88 does not seem impressive.

Recall that the prediction task is nondeterministic. Successors cannot be predicted with absolute certainty; there is a built-in error which is inevitable. Nevertheless, although the prediction cannot be error-free, it is also true that word order is not random. For any given sequence of words there are a limited number of possible successors. Under these circumstances, the network should learn the expected frequency of occurrence of each of the possible successor words; it should then activate the output nodes proportional to these expected frequencies.

This suggests that rather than testing network performance with the RMS error calculated on the actual successors, the output should be compared

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with the expected frequencies of occurrence of possible successors. These expected latter values can be determined empirically from the training corpus. Every word in a sentence is compared against all other sentences that are, up to that point, identical. These constitute the comparison set. The probability of occurrence for all possible successors is then determined from this set. This yields a vector for each word in the training set. The vector is of the same dimensionality as the output vector, but rather than representing a distinct word (by turning on a single bit), it represents the likelihood of each possible word occurring next (where each bit position is a fractional number equal to the probability). For testing purposes, this likelihood vector can be used in place of the actual teacher and a RMS error computed based on the comparison with the network output. (Note that it is appropriate to use these likelihood vectors only for the testing phase. Training must be done on actual successors, because the point is to force the network to learn the probabilities.)

When performance is evaluated in this manner, RMS error on the training set is 0.053 (SD = 0.100). One remaining minor problem with this error measure is that although the elements in the likelihood vectors must sum to 1.0 (since they represent probabilities), the activations of the network need not sum to 1.0. It is conceivable that the network output learns the relative frequency of occurrence of successor words more readily than it approximates exact probabilities. In this case the shape of the two vectors might be similar, but their length different. An alternative measure which normalizes for length differences and captures the degree to which the shape of the vectors is similar is the cosine of the angle between them. Two vectors might be parallel (cosine of 1.0) but still yield an RMS error, and in this case it might be felt that the network has extracted the crucial information. The mean cosine of the angle between network output on training items and likelihood vectors is 0.916 (SD = 0.123). By either measure, RMS or cosine, the network seems to have learned to approximate the likelihood ratios of potential successors.

How has this been accomplished? The input representations give no information (such as form class) that could be used for prediction. The word vectors are orthogonal to each other. Whatever generalizations are true of classes of words must be learned from the co-occurrence statistics, and the composition of those classes must itself be learned.

If indeed the network has extracted such generalizations, as opposed simply to memorizing the sequence, one might expect to see these patterns emerge in the internal representations which the network develops in the course of learning the task. These internal representations are captured by the pattern of hidden unit activations which are evoked in response to each word and its context. (Recall that hidden units are activated by both input units and context units. There are no representations of words in isolation.)

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The nature of these internal representations was studied in the following way. After the learning phase of six complete passes through the corpus, the connection strengths in the network were frozen. The input stream was passed through the network one final time, with no learning taking place. During this testing, the network produced predictions of future inputs on the output layer. These were ignored. Instead, the hidden unit activations for each word + context input were saved, resulting in 27,354 150-bit vectors. Each word occurs many times, in different contexts. As a first approximation of a word's prototypical or composite representation, all hidden unit activation patterns produced by a given word (in all its contexts) were averaged to yield a single 150-bit vector for each of the 29 unique words in the input stream.[6] (In the next section it will be shown how it is possible to study the internal representations of words in context.) These internal representations were then subject to a hierarchical clustering analysis. Figure 7 shows the resulting tree; this tree reflects the similarity structure of the internal representations these lexical items. Lexical items which have similar properties are grouped together lower in the tree, and clusters of similar words which resemble other clusters are connected higher in the tree.

The network has discovered that there are several major categories of words. One large category corresponds to *verbs;* another category corresponds to *nouns.* The verb category is broken down into groups that *require a direct object,* or are *intransitive,* or where a *direct object is optional.* The noun category is broken into two major groups: *inanimates,* and *animates.* Animates are divided into *human* and *nonhuman;* the nonhuman are divided into *large animals* and *small animals.* Inanimates are broken into *breakable, edibles,* and nouns which appeared as subjects of agentless active verbs.

The network has developed internal representations for the input vectors which reflect facts about the possible sequential ordering of the inputs. The network is not able to predict the precise order of words, but it recognizes that (in this corpus) there is a class of inputs (namely, verbs) that typically follow other inputs (namely, nouns). This knowledge of class behavior is quite detailed; from the fact that there is a class of items which always precedes "chase," "break," "smash," it infers that the large animals form a class.

[6] Tony Plate (personal communication) has pointed out that this technique is dangerous, inasmuch as it may introduce a statistical artifact. The hidden unit activation patterns are highly dependent upon preceding inputs. Because the preceding inputs are not uniformly distributed (they follow precisely the co-occurrence conditions which are appropriate for the different categories), this means that the mean hidden unit pattern across all contexts of a specific item will closely resemble the mean hidden unit pattern for other items in the same category. This could occur even without learning, and is a consequence of the averaging of vectors which occurs prior to cluster analysis. Thus the results of the averaging technique should be verified by clustering individual tokens; tokens should always be closer to other members of the same type than to tokens of other types.

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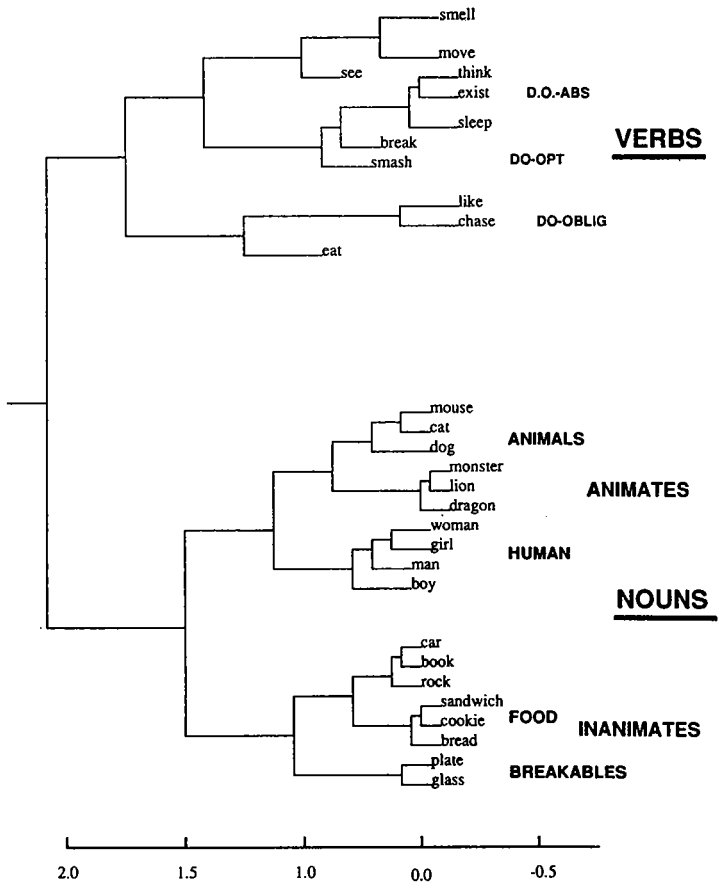


Figure 7. Hierarchical cluster diagram of hidden unit activation vectors in simple sentence prediction task. Labels indicate the inputs which produced the hidden unit vectors; inputs were presented in context, and the hidden unit vectors averaged across multiple contexts.

Several points should be emphasized. First, the category structure appears to be hierarchical. Thus, "dragons" are large animals, but also members of the class of [−human, +animate] nouns. The hierarchical interpretation is achieved through the way in which the spatial relations (of the representations) are organized. Representations that are near one another in the representational space form classes, while higher level categories correspond to larger and more general regions of this space.

Second, it is also true that the hierarchy is "soft" and implicit. While some categories may be qualitatively distinct (i.e., very far from each other

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in space), there may also be other categories that share properties and have less distinct boundaries. Category membership in some cases may be marginal of unambiguous.

Finally, the content of the categories is not known to the network. The network has no information available which would "ground" the structural information in the real world. In this respect, the network has much less information to work with than is available to real language learners.[7] In a more realistic model of acquisition, one might imagine that the utterance provides one source of information about the nature of lexical categories; the world itself provides another source. One might model this by embedding the "linguistic" task in an environment; the network would have the dual task of extracting structural information contained in the utterance, and structural information about the environment. Lexical meaning would grow out of the associations of these two types of input.

In this simulation, an important component of meaning is context. The representation of a word is closely tied up with the sequence in which it is embedded. Indeed, it is incorrect to speak of the hidden unit patterns as word representations in the conventional sense, since these patterns also reflect the prior context. This view of word meaning, that is, its dependence upon context, can be demonstrated in the following way.

Freeze the connections in the network that has just been trained, so that no further learning occurs. Imagine a novel word, *zog,* which the network has never seen before, and assign to this word a bit pattern which is different from those it was trained on. This word will be used in place of the word *man;* everywhere that *man* could occur, *zog* will occur instead. A new sequence of 10,000 sentences is created, and presented once to the trained network. The hidden unit activations are saved, and subjected to a hierarchical clustering analysis of the same sort used with the training data.

The resulting tree is shown in Figure 8. The internal representation for the word *zog* bears the same relationship to the other words as did the word *man* in the original training set. This new word has been assigned an internal representation that is consistent with what the network has already learned (no learning occurs in this simulation) and the new word's behavior. Another way of looking at this is in certain contexts, the network expects *man,* or something very much like it. In just such a way, one can imagine real language learners making use of the cues provided by word order to make intelligent guesses about the meaning of novel words.

Although this simulation was not designed to provide a model of context effects in word recognition, its behavior is consistent with findings that have been described in the experimental literature. A number of investigators

[7] Jay McClelland has suggested a humorous—but entirely accurate—metaphor for this task: It is like trying to learn a language by listening to the radio.

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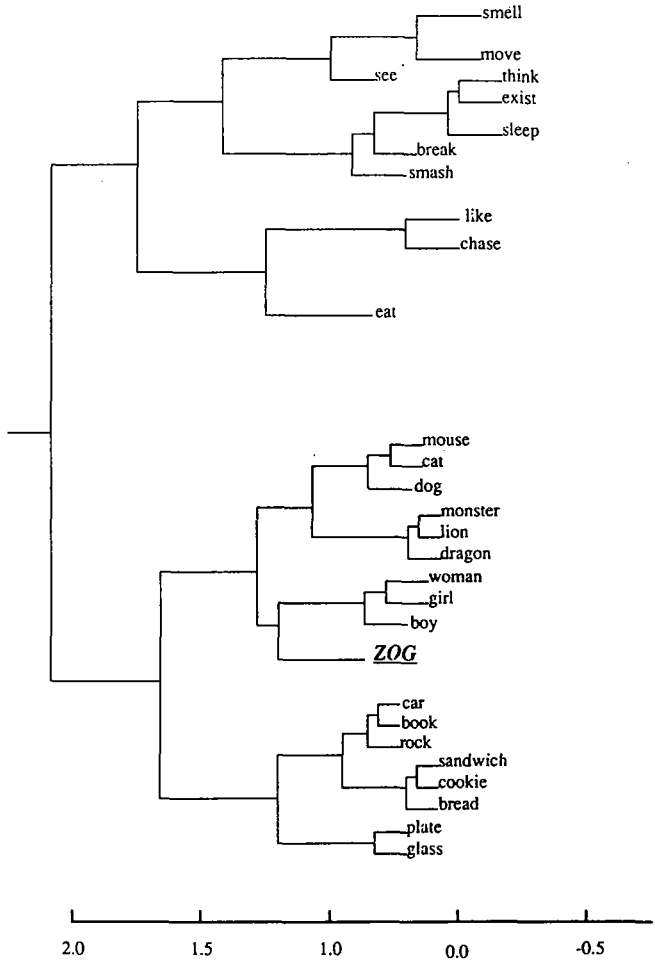


Figure 8. Hierarchical clustering diagram of hidden unit activation vectors in simple sentence prediction task, with the addition of the novel input ZOG.

have studied the effects of sentential context on word recognition. Although some researchers have claimed that lexical access is insensitive to context (Swinney, 1979), there are other results which suggest that when context is sufficiently strong, it does indeed selectively facilitate access to related words (Tabossi, Colombo, & Job, 1987). Furthermore, individual items are typically not very predictable but classes of words are (Schwanenflugel & Shoben, 1985; Tabossi, 1988). This is precisely the pattern found here, in which the error in predicting the actual next word in a given context remains high, but

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the network is able to predict the approximate likelihood of occurrence of classes of words.

TYPES, TOKENS, AND STRUCTURED REPRESENTATIONS

There has been considerable discussion about the ways in which PDP networks differ from traditional computational models. One apparent difference is that traditional models involve symbolic representations, whereas PDP nets seem, to many people, to be non- or perhaps subsymbolic (Fodor & Pylyshyn, 1988; Smolensky, 1987, 1988). This is a difficult and complex issue, in part because the definition of symbol is problematic. Symbols do many things, and it might be more useful to contrast PDP versus traditional models with regard to the various functions that symbols can serve.

Both traditional and PDP networks involve representations which are symbolic in the specific sense that the representations refer to other things. In traditional systems, the symbols have names such as A, or x, or β. In PDP nets, the internal representations are generally activation patterns across a set of hidden units. Although both kinds of representations do the task of referring, there are important differences. Classical symbols typically refer to classes or categories, but in PDP nets the representations may be highly context-dependent. This does not mean that the representations do not capture information about category or class (this should be clear from the previous simulation); it does mean that there is also room in the representation scheme to pick out individuals.

This property of PDP representations might seem to be a serious drawback to some. In the extreme, it suggests that there could be separate representations for the entity *John* in every different context in which that entity can occur, leading to an infinite number of $John_i$. But rather than being a drawback, I suggest this aspect of PDP networks significantly extends their representational power. The use of distributed representations, together with the use of context in representing words (which is a consequence of simple recurrent networks) provides one solution to a thorny problem—the question of how to represent type/token differences—and sheds insight on the ways in which distributed representations can represent structure.

In order to justify this claim, let me begin by commenting on the representational richness provided by the distributed representations developed across the hidden units. In localist schemes, each node stands for a separate concept. Acquiring new concepts usually requires adding new nodes. In contrast, the hidden unit patterns in the simulations reported here have tended to develop distributed representations. In this scheme, concepts are expressed as activation patterns over a fixed number of nodes. A given node participates in representing multiple concepts. It is the activation pattern in its entirety that is meaningful. The activation of an individual node may be

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uninterpretable in isolation (i.e., it may not even refer to a feature or microfeature).

Distributed representations have a number fo advantages over localist representations (although the latter are not without their own benefits).[8] If the units are analog (i.e., capable of assuming activation states in a continuous range between some minimum and maximum values), then, in principle, there is no limit to the number of concepts which can be represented with a finite set of units. In the simulations here, the hidden unit patterns do double duty. They are required not only to represent inputs, but to develop representations which will serve as useful encodings of temporal context that can be used when processing subsequent inputs. Thus, in theory, analog hidden units would also be capable of providing infinite memory.

Of course, there are many reasons why in practice the memory is bounded, and why the number of concepts that can be stored is finite. There is limited numeric precision in the machines on which these simulations are run; the activation function is repetitively applied to the memory and results in exponential decay; and the training regimen may not be optimal for exploiting the full capacity of the networks. For instance, many of the simulations reported here involve the prediction task. This task incorporates feedback on every training cycle. In other pilot work, it was found that there was poorer performance in tasks in which there was a delay in injecting error into the network. Still, just what the representational capacity is of these simple recurrent networks remains an open question (but, see Servan-Schreiber, Cleeremans, & McClelland, 1988).

Having made these preliminary observations, the question of the context-sensitivity of the representations developed in the simulations reported here will be addressed. Consider the sentence-processing simulation. It was found that after learning to predict words in sentence sequences, the network developed representations that reflected aspects of the words' meaning as well as their grammatical category. This was apparent in the similarity structure of the internal representation of each word; this structure was presented graphically as a tree in Figure 7.

In what sense are the representations, which have been clustered in Figure 7, context sensitive? In fact, they are not; recall that these representations are composites of the hidden unit activation patterns in response to each word averaged across many different contexts. So the hidden unit activation pattern used to represent *boy,* for instance, was really the mean vector of activation patterns in response to *boy* as it occurs in many different contexts.

The reason for using the mean vector in the previous analysis was in large part practical. It is difficult to do a hierarchical clustering of 27,454 patterns, and even more difficult to display the resulting tree graphically. However,

[8] These advantages are discussed at length in Hinton, McClelland, and Rumelhart (1986).

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one might want to know whether the patterns displayed in the tree in Figure 7 are in any way artifactual. Thus, a second analysis was carried out, in which all 27,454 patterns were clustered. The tree cannot be displayed here, but the numerical results indicate that the tree would be identical to the tree shown in Figure 7; except that instead of ending with the terminals that stand for the different lexical items, the branches would continue with further arborization containing the specific instances of each lexical item in its context. No instance of any lexical item appears inappropriately in a branch belonging to another.

It would be correct to think of the tree in Figure 7 as showing that the network has discovered that there are 29 *types* (among the sequence of 27,454 inputs). These types are the different lexical items shown in that figure. A finer grained analysis reveals that the network also distinguishes between the specific occurrences of each lexical item, that is, the *tokens*. The internal representations of the various tokens of a lexical type are very similar. Hence, they are all gathered under a single branch in the tree. However, the internal representations also make subtle distinctions between (for example), *boy* in one context and *boy* in another. Indeed, as similar as the representations of the various tokens are, no two tokens of a type are exactly identical.

Even more interesting is that there is a substructure of the representations of the various types of a token. This can be seen by looking at Figure 9, which shows the subtrees corresponding to the tokens of *boy* and *girl*. (Think of these as expansions of the terminal leaves for *boy* and *girl* in Figure 8.) The individual tokens are distinguished by labels which indicate their original context.

One thing that is apparent is that subtrees of both types (*boy* and *girl*) are similar to one another. On closer scrutiny, it is seen that there is some organization here; (with some exceptions) tokens of *boy* that occur in sentence-initial position are clustered together, and tokens of *boy* in sentence-final position are clustered together. Furthermore, this same pattern occurs among the patterns representing *girl*. Sentence-final words are clustered together on the basis of similarities in the preceding words. The basis for clustering of sentence-initial inputs is simply that they are all preceded by what is effectively noise (prior sentences). This is because there are no useful expectations about the sentence-initial noun (other than that it will be a noun) based upon the prior sentences. On the other hand, one can imagine that if there were some discourse structure relating sentences to each other, then there might be useful information from one sentence which would affect the representation of sentence-initial words. For example, such information might disambiguate (i.e., give referential content to) sentence-initial pronouns.

Once again, it is useful to try to understand these results in geometric terms. The hidden unit activation patterns pick out points in a high (but fixed) dimensional space. This is the space available to the network for its

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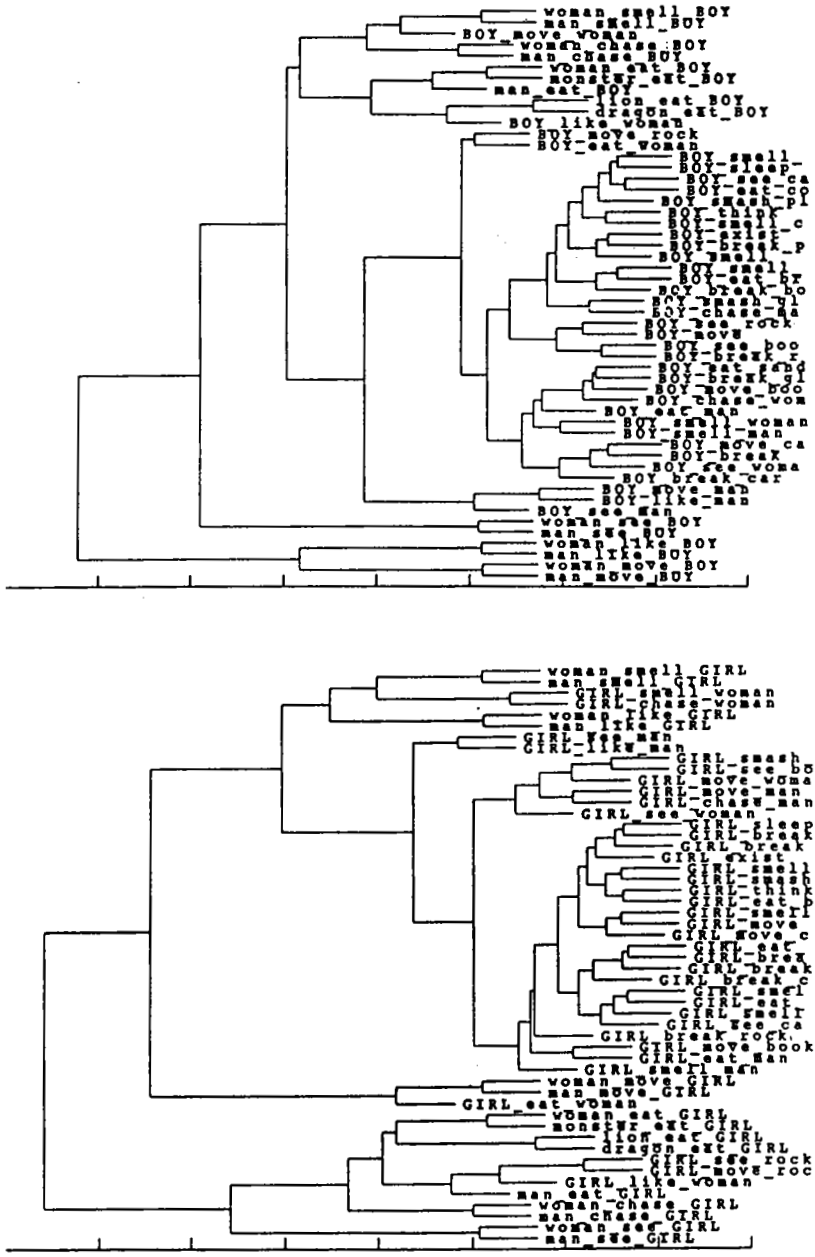


Figure 9. Hierarchical cluster diagram of hidden unit activation vectors in response to some occurrences of the inputs BOY and GIRL. Upper-case labels indicate the actual input; lower-case labels indicate the context for each input.

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internal representations. The network structures that space in such a way that important relations between entities is translated into spatial relationships. Entities which are nouns are located in one region of space and verbs in another. In a similar manner, different types (here, lexical items) are distinguished from one another by occupying different regions of space; but also, tokens of a same type are differentiated. The differentiation is nonrandom, and the way in which tokens of one type are elaborated is similar to elaboration of another type. That is, $John_1$ bears the same spatial relationtionship to $John_2$ as $Mary_1$ bears to $Mary_2$.

This use of context is appealing, because it provides the basis both for establishing generalizations about classes of items and also allows for the tagging of individual items by their context. The result is that types can be identified at the same time as tokens. In symbolic systems, type/token distinctions are often made by indexing or binding operations; the networks here provide an alternative account of how such distinctions can be made without indexing or binding.

CONCLUSIONS

There are many human behaviors which unfold over time. It would be folly to try to understand those behaviors without taking into account their temporal nature. The current set of simulations explores the consequences of attempting to develop representations of time that are distributed, task-dependent, and in which time is represented implicitly in the network dynamics.

The approach described here employs a simple architecture, but is surprisingly powerful. There are several points worth highlighting.

- *Some problems change their nature when expressed as temporal events.* In the first simulation, a sequential version of the XOR was learned. The solution to this problem involved detection of state changes, and the development of frequency-sensitive hidden units. Casting the XOR problem in temporal terms led to a different solution than is typically obtained in feed-forward (simultaneous input) networks.
- *The time-varying error signal can be used as a clue to temporal structure.* Temporal sequences are not always uniformly structured, nor uniformly predictable. Even when the network has successfully learned about the structure of a temporal sequence, the error may vary. The error signal is a good metric of where structure exists; it thus provides a potentially very useful form of feedback to the system.
- *Increasing the sequential dependencies in a task does not necessarily result in worse performance.* In the second simulation, the task was complicated by increasing the dimensionality of the input vector, by extending the duration of the sequence, and by making the duration of the sequence variable. Performance remained good, because these com-

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plications were accompanied by redundancy, which provided additional cues for the task. The network was also able to discover which parts of the complex input were predictable, making it possible to maximize performance in the face of partial unpredictability.

- *The representation of time—and memory—is highly task-dependent.* The networks here depend upon internal representations which have available, as part of their input, their own previous state. In this way the internal representations intermix the demands of the task with the demands imposed by carrying out that task over time. There is no separate "representation of time." There is simply the representation of input patterns in the context of a given output function; it just happens that those input patterns are sequential. That representation, and thus the representation of time, varies from task to task. This presents a somewhat novel view of memory. In this account, memory is neither passive nor a separate subsystem. One cannot properly speak of a memory for sequences; that memory is inextricably bound up with the rest of the processing mechanism.
- *The representations need not be "flat," atomistic, or unstructured.* The sentence task demonstrated that sequential inputs may give rise to internal representations which are hierarchical in nature. The hierarchy is implicit in the similarity structure of the hidden unit activations and does not require an a priori architectural commitment to the depth or form of the hierarchy. Importantly, distributed representations make available a space which can be richly structured. Categorical relationships as well as type/token distinctions are readily apparent. Every item may have its own representation, but because the representations are structured, relations between representations are preserved.

The results described here are preliminary in nature. They are highly suggestive, and often raise more questions than they answer. These networks are properly thought of as dynamical systems, and one would like to know more about their properties as such. For instance, the analyses reported here made frequent use of hierarchical clustering techniques in order to examine the similarity structure of the internal representations. These representations are snapshots of the internal states during the course of processing a sequential input. Hierarchical clustering of these snapshots gives useful information about the ways in which the internal states of the network at different points in time are similar or dissimilar. But the temporal relationship between states is lost. One would like to know what the trajectories between states (i.e., the vector field) look like. What sort of attractors develop in these systems? It is a problem, of course, that the networks studied here are high-dimensional systems, and consequently difficult to study using traditional techniques. One promising approach, which is currently being studied, is to carry out a principal components analysis of the hidden unit

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activation pattern time series, and then to construct phase state portraits of the most significant principal components (Elman, 1989).

Another question of interest is what is the memory capacity of such networks. The results reported here suggest that these networks have considerable representational power; but more systematic analysis using better defined tasks is clearly desirable. Experiments are currently underway using sequences generated by finite state automata of various types; these devices are relatively well understood, and their memory requirements may be precisely controlled (Servan-Schreiber et al., 1988).

One of the things which feedforward PDP models have shown is that simple networks are capable of discovering useful and interesting internal representations of many static tasks. Or put the other way around: Rich representations are implicit in many tasks. However, many of the most interesting human behaviors have a serial component. What is exciting about the present results is that they suggest that the inductive power of the PDP approach can be used to discover structure and representations in tasks which unfold over time.

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